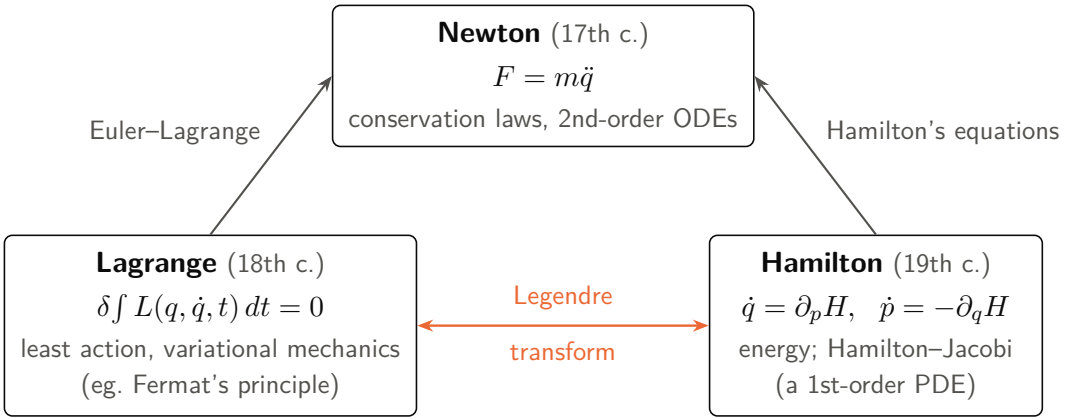
Extremal metrics on polygons

Brian Weber

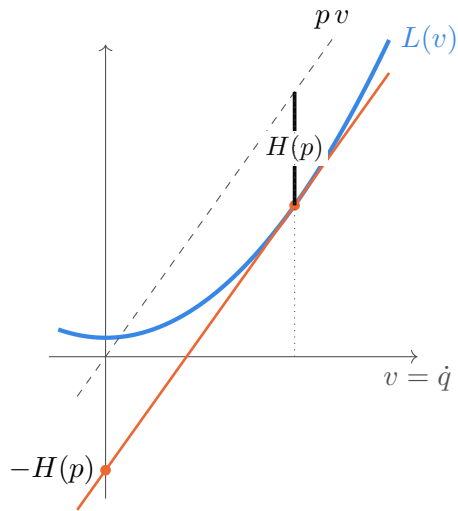
Three pictures of mechanics



The Legendre transform

$$H(q,p,t) \;=\; \dot{q}_i\, p^i - L(q,\dot{q},t), \qquad p^i = \frac{\partial L}{\partial \dot{q}_i}$$

- velocities $\dot{q} \;\longleftrightarrow\;$ momenta p
- $H(p) = \max_v \left(p\, v - L(v)\right)$
- convex $\leftrightarrow$ convex. Involution: doing it twice gives back L



Underneath: symplectic geometry (20th c.)

A **symplectic form** on a manifold M^n is a 2-form ω with

$$d\omega = 0, \qquad \underbrace{\omega \wedge \cdots \wedge \omega}_{n/2} \neq 0.$$

- Classically $\omega = dq_i \wedge dp^i$ (e.g. Courant–Hilbert, 1924)
- The *geometry* is ω; the *physics* is a choice of Hamiltonian H:

$$\omega(X_H, \cdot\,) = dH \quad \rightsquigarrow \quad \text{the flow of } X_H \text{ is the action (the kinematic flow)}$$

Phase space

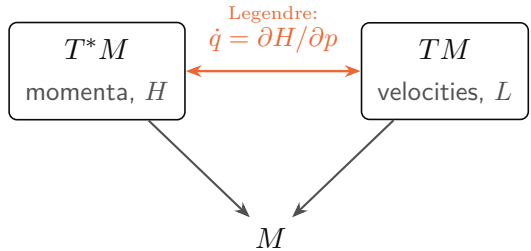
- The cotangent bundle T^*M carries a natural (tautological) symplectic form

$$\theta = p^i \, dq_i, \qquad \omega = d\theta.$$

Invariant form: $v \in TT^*M, (p,q) \in T^*M, \pi : T^*M \to M$, then

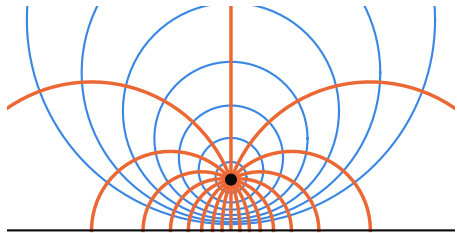
$$\theta_{(q,p)}(v) = p(\pi_* v), \theta \in T^*T^*M$$
$$\omega = d\theta \in \bigwedge{}^2 T^*M$$

- A Hamiltonian (or a Lagrangian) creates a Legendre transform. This is naturally a linear (on fibers) isomorphism between T^*M and TM.



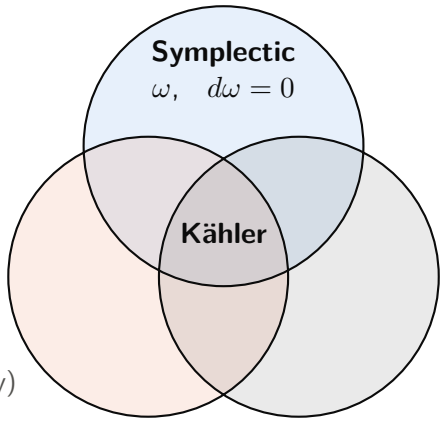
A metric is a Hamiltonian

- Kinetic energy $H = \frac{1}{2}\, g^{ij} p_i p_j$.
  The metric g is the *polarization* of this quadratic form.
- Its Legendre map $T^*M \to TM$ is raising and lowering indices.
- Geodesic flow $=$ the motion of a free particle.
  Wavefronts solve Hamilton–Jacobi: $|\nabla S|_g = 1$.



hyperbolic plane: geodesics from one point, wavefronts

Three geometries



Complex
$J, \quad J^2 = -1$
(incl. algebraic geometry)

Riemannian
g

$$g + \omega(J\cdot, \cdot) = 0$$

$$U(n) = O(2n) \cap Sp(2n, \mathbb{R}) \cap GL(n, \mathbb{C})$$

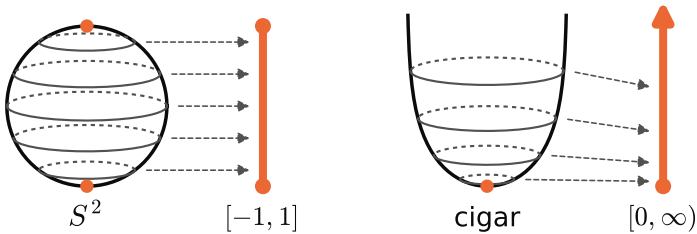
Symmetry

- Every symmetry gives a conserved quantity: a **momentum** (Noether).
- Phase space of dimension $2n$ with n commuting symmetries:
  the motion is *determined*. (Liouville–Arnold: action–angle variables)

- Kähler manifold of dimension $2n$ with n commuting symmetries (a torus T^n):
  the geometry is determined by the momenta.

- The momenta together form the **moment map** $\mu : M^{2n} \to \mathbb{R}^n$.

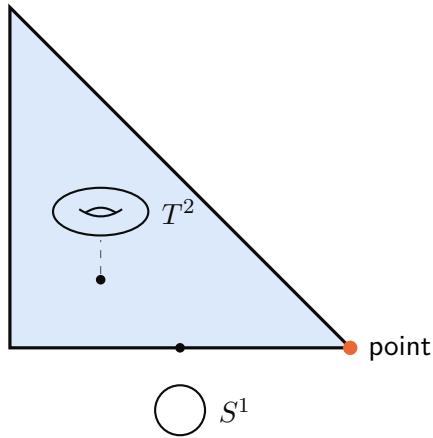
Moment maps



rotation about the axis; the moment map is the height. Orbits are circles, which collapse to points at the ends.

In dimension 4: a polygon

- T^2 acts on M^4; $\mu : M^4 \to \mathbb{R}^2$.
- The image is a convex **polygon**.
- Over each point: the orbit.
  - interior: a torus T^2
  - edge: a circle (one circle collapses)
  - vertex: a point
- Coordinates: momenta $x \in P$ and angles $\theta \in T^2$.

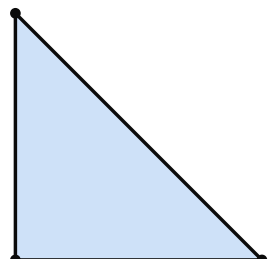


Delzant polygons

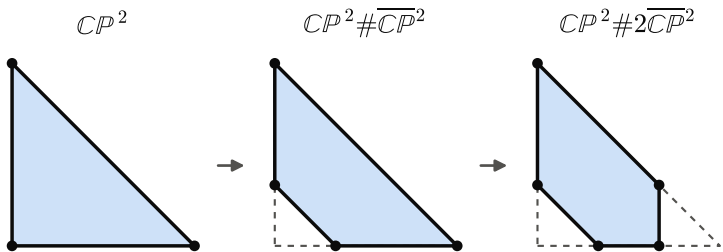
- **Delzant** (1988): compact toric symplectic manifolds $\leftrightarrow$ polygons whose edges at each vertex form a basis of $\mathbb{Z}^2$.
- $\mathbb{CP}^2$, with T^2 acting by $[z_0 : z_1 : z_2] \mapsto [z_0 : e^{i\theta_1} z_1 : e^{i\theta_2} z_2]$:

$$\mu = \left(\frac{|z_1|^2}{|z_0|^2 + |z_1|^2 + |z_2|^2}, \frac{|z_2|^2}{|z_0|^2 + |z_1|^2 + |z_2|^2} \right).$$

The image is a triangle.



Blowing up = cutting off a corner



- One blow-up: the **Page** Einstein metric is conformal to an extremal Kähler metric on it.
- Two blow-ups: so is the **Chen–LeBrun–Weber** Einstein metric.

The same polygon, another way

$\mathbb{CP}^1 \times \mathbb{CP}^1$ $\qquad$ $(\mathbb{CP}^1 \times \mathbb{CP}^1)\#\overline{\mathbb{CP}}^2$



$(\mathbb{CP}^1 \times \mathbb{CP}^1)\#\overline{\mathbb{CP}}^2 \cong \mathbb{CP}^2\#2\overline{\mathbb{CP}}^2$: cutting one corner of the square gives the same pentagon.

Geometry on the polygon: the symplectic potential

- One convex function u on P gives the metric

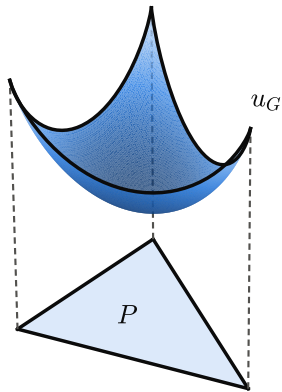
$$g = u_{ij}\, dx^i dx^j + u^{ij}\, d\theta_i d\theta_j.$$

- Edges $\ell_k(x) = \langle u_k, x\rangle - \lambda_k \geq 0$.
- Guillemin's potential

$$u_G = \tfrac{1}{2} \sum_k \ell_k \log \ell_k$$

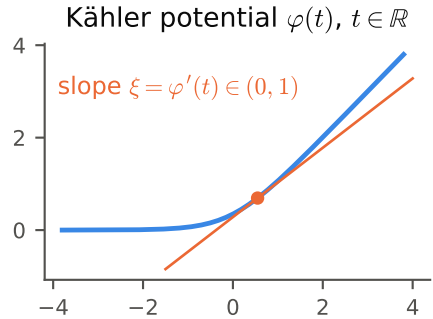
with log-singularities at the boundary.

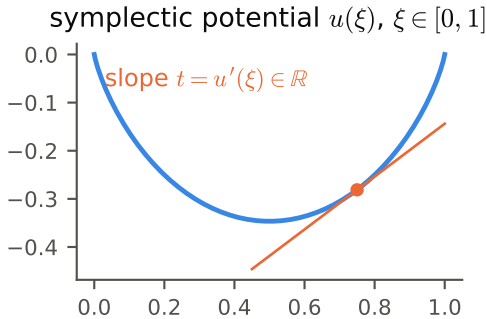
- Every toric Kähler metric: $u = u_G +$ smooth (Abreu).



u_G as a graph over the $\mathbb{CP}^2$ triangle P

The Legendre transform, again





$$\varphi(t) + u(\xi) = \langle t, \xi \rangle, \qquad \xi = \nabla\varphi(t), \qquad t = \nabla u(\xi)$$

φ: Kähler potential in log-coordinates $t = \log|z|$ $\quad u$: symplectic potential on the polygon

The fundamental duality

	mechanics	**toric Kähler geometry**
one side	velocities $\dot{q}$, Lagrangian L	complex coordinates t, Kähler potential φ
other side	momenta p, Hamiltonian H	momenta $\xi \in P$, symplectic potential u
relation	$p = \partial L / \partial \dot{q}$	$\xi = \nabla \varphi$
geometry	Lagrange $\leftrightarrow$ Hamilton	complex $\leftrightarrow$ symplectic

Extremal Kähler metrics

- Fix the Kähler class. **Calabi functional**:

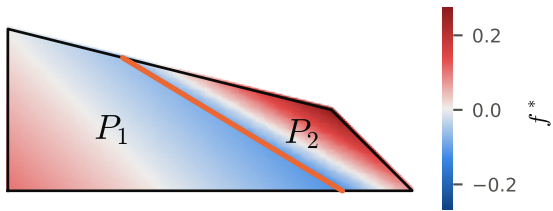
$$\mathrm{Cal}(g) = \int_M S^2 \, d\mu \qquad (S = \text{scalar curvature})$$

- Its critical points are the **extremal** metrics: ∇S is a holomorphic vector field.
- Examples:
  - constant scalar curvature (cscK)
  - Kähler–Einstein: $\mathrm{Ric} = \lambda g$
  - general extremal metrics

- Toric: $S = -\frac{1}{2}\,\partial_i \partial_j u^{ij}$ (Abreu). Extremal $\iff$ S is an affine function on P.

Geometrization

Conjecture. A Kähler manifold has an essentially unique decomposition into pieces, each of which carries an extremal Kähler metric. (cf. Donaldson; Székelyhidi)

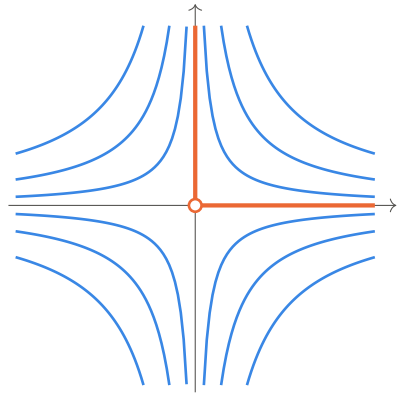


a polygon with no extremal metric: the best one can do breaks it along a line into P_1, P_2

Obstructions: stability

- "Stability" comes from geometric invariant theory (Hilbert, Mumford).
- To form a quotient by a group, keep only orbits that are **closed**.
- Test with one-parameter subgroups: a sign of a weight.

- Here: instability is an obstruction to cscK and extremal metrics.



hyperbolas: closed orbits (stable)

axes: the limit 0 is not in the orbit

$$\lambda \cdot (x, y) = (\lambda x, \lambda^{-1} y)$$

The conjecture

Relatively K-polystable $\iff$ an extremal metric exists

(Yau–Tian–Donaldson; Székelyhidi for extremal metrics)

- Open in general.
- A theorem for toric surfaces, i.e. for polygons (Donaldson; Chen–Li–Sheng).
  So polygons are a laboratory where stability can be *computed*.

The obstruction on a polygon

Donaldson's linear functional, for f convex on P:

$$\mathcal{L}_A(f) \;=\; \int_{\partial P} f\, d\sigma \;-\; \int_P A\, f\, d\mu$$

- $A = A_{\mathrm{ext}}$: the affine function with $\mathcal{L}_A(\text{affine}) = 0$ (computable from P).
- If $S(u) = A$, integrating by parts gives

$$\mathcal{L}_A(f) = \tfrac{1}{2} \int_P u^{ij} f_{ij}\, d\mu \;\geq\; 0.$$

- **Stability**: $\mathcal{L}_A(f) > 0$ for every convex, non-affine f.
- One f with $\mathcal{L}_A(f) < 0$ proves that no extremal metric exists.

The nonlinear functional: Donaldson–Mabuchi

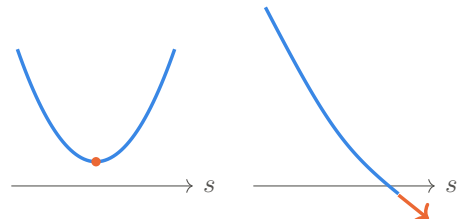
$$\mathcal{F}(u) \;=\; -\int_P \log\det(\mathrm{Hess}\, u)\, d\mu \;+\; 2\,\mathcal{L}_A(u)$$

Mabuchi's K-energy, written on the polygon (Donaldson)

- $\mathcal{F}$ is **convex** on symplectic potentials $u = u_G +$ smooth.
- $d\mathcal{F}_u(f) = 2\int_P (S(u) - A) f\, d\mu$: critical points = extremal metrics.
- $\mathcal{L}_A$ is the slope at infinity:

$$\mathcal{F}(u + sf) \;\le\; \mathcal{F}(u) + 2s\,\mathcal{L}_A(f).$$

- $\mathcal{L}_A(f) < 0 \Rightarrow \mathcal{F}$ is unbounded below $\Rightarrow$ no minimizer.



stable: a minimum

unstable: slope $2\mathcal{L}_A(f) < 0$

$\mathcal{F}$ along a ray $u + sf$

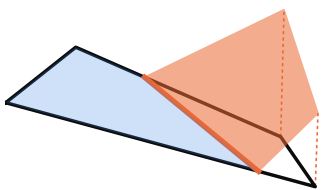
Crease functions

- In dimension 2 it suffices to test **simple creases**

$$f = \max(0,\ \ell), \quad \ell \text{ affine}$$

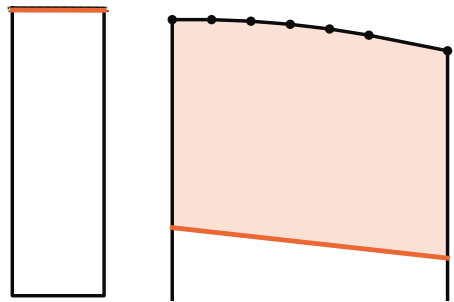
(Wang–Zhou)

- One line in the polygon.
- Instability = a line with $\mathcal{L}_A(f) < 0$.

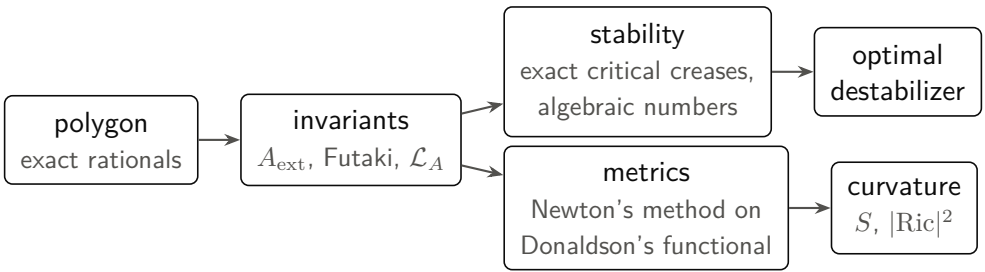


Example: the Wang–Zhou nonagon

- A Delzant polygon: a *smooth* toric surface.
- No extremal metric in this class.
- Destabilizing line
  $x + 0.358\, y = 3537.9$
  (height parameter $\alpha = 10^4$; found by our code)



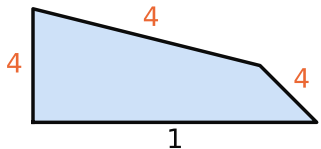
The code



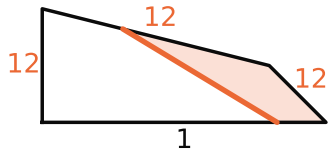
- Approximate extremal metrics: minimize the Donaldson–Mabuchi functional $\mathcal{F}$ over $u = u_G +$ polynomial.
- Python; `github.com/lapisetanima/toric-kahlers`

Orbifolds: easier to destabilize

extremal metric exists

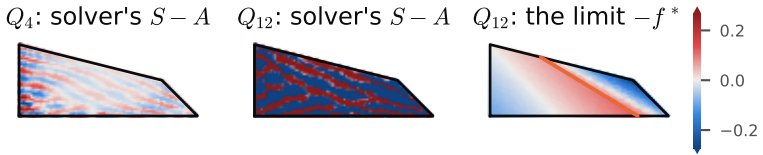


unstable



- A **label** m on an edge: cone angle $2\pi/m$ along that divisor.
- Same quadrilateral, labels $(1, 4, 4, 4)$ and $(1, 12, 12, 12)$: one has an extremal metric, the other does not.

When there is no metric, the approximation goes haywire

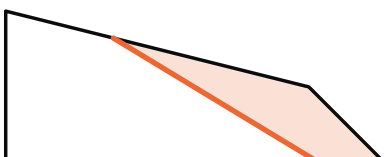


degree	Q_4	Q_{12}
10	0.21	$1.0 \cdot 10^3$
14	0.064	10
18	0.022	$4.6 \cdot 10^3$

$\|S - A\|_{L^2}$ (area-normalized). For Q_{12} it is at least 0.064.

- Q_4: the approximations converge.
- Q_{12}: they cannot. A *global* polynomial can't break along the line.
- The right answer is piecewise: $S \to A - f^*$.

Exact answers

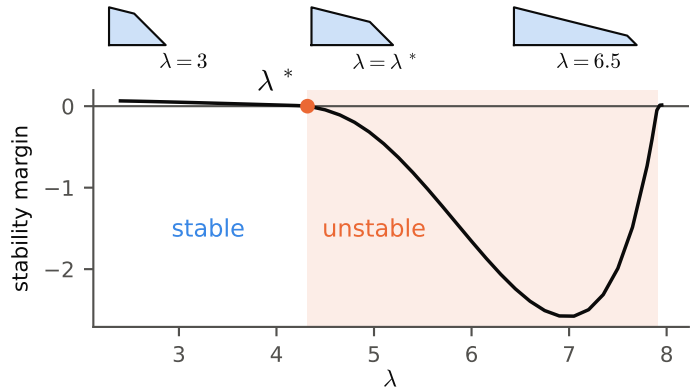


Optimal destabilizer of Q_{12}:

- line $x + m\,y = s$,
  $m = 1.65238288\ldots$
- m is a root of an integer polynomial of degree 18
- $\|f^*\| =$
  $0.16417411137442506876\ldots$

  a lower bound for $\|S - A_{\text{ext}}\|$ over all toric metrics

A threshold

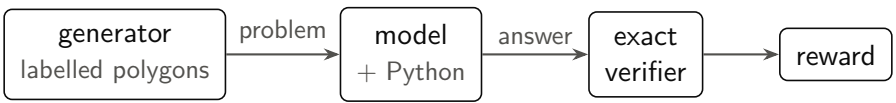


$$\lambda^* = 4.31756909532834813881399236429256191048568\ldots$$

exact critical creases; confirmed independently by Apostolov–Calderbank–Gauduchon's criterion for quadrilaterals

A testbed for AI

- Answers are **exactly checkable**: a destabilizing line is a certificate.
- Problems are cheap to generate, with tunable difficulty.
- Disguised: no mention of Kähler geometry; the model gets a Python tool.



Tasks (work in progress)

- **Classify**: stable or unstable?
- **Certify**: produce a line with $\mathcal{L}_A < 0$. Checked exactly.
- **Optimize**: the optimal line, to as many digits as possible. Reward $\sim$ digits correct.

- Quadrilaterals have a closed-form theory (Apostolov–Calderbank–Gauduchon): too easy to look up.
- Pentagons and beyond: no closed form. This is where models must discover something.

Thank you